Thaddeus Harris	Materials for the Built Env.	Homework #10 10/16/2024

(2) Carbon increases the brittleness of the steel, and is mechanically stronger, whereas milder steel is weaker and more ductile. Generally hardness increases and ductility decreases with increased carbon content.

(5) A 5ft section of HP14x102 weighs 510 lbs

(7) Axial load of 200 N, determine stress & deformation of an aluminum wire 20m long & 5mm wide.

$2.5mm^2 \pi = A = 19.63mm^2$

$E = 70GPa \quad \sigma = \frac{200N}{19.63mm^2} = 10.18N/mm^2 = \underline{\underline{1.02\times10^{-5} N/m^2}}$

$\varepsilon = \frac{1.02\times10^{-5} N/m^2}{70GPa} = 1.46\times10^{-13}$

$disp. = 1.46\times10^{-13} \times 20 = \underline{\underline{2.91\times10^{-12}}}$

(11) $\varepsilon = \frac{\sigma}{50}\left[1+\frac{3}{7}\left(\frac{\sigma}{50}\right)^5\right]$

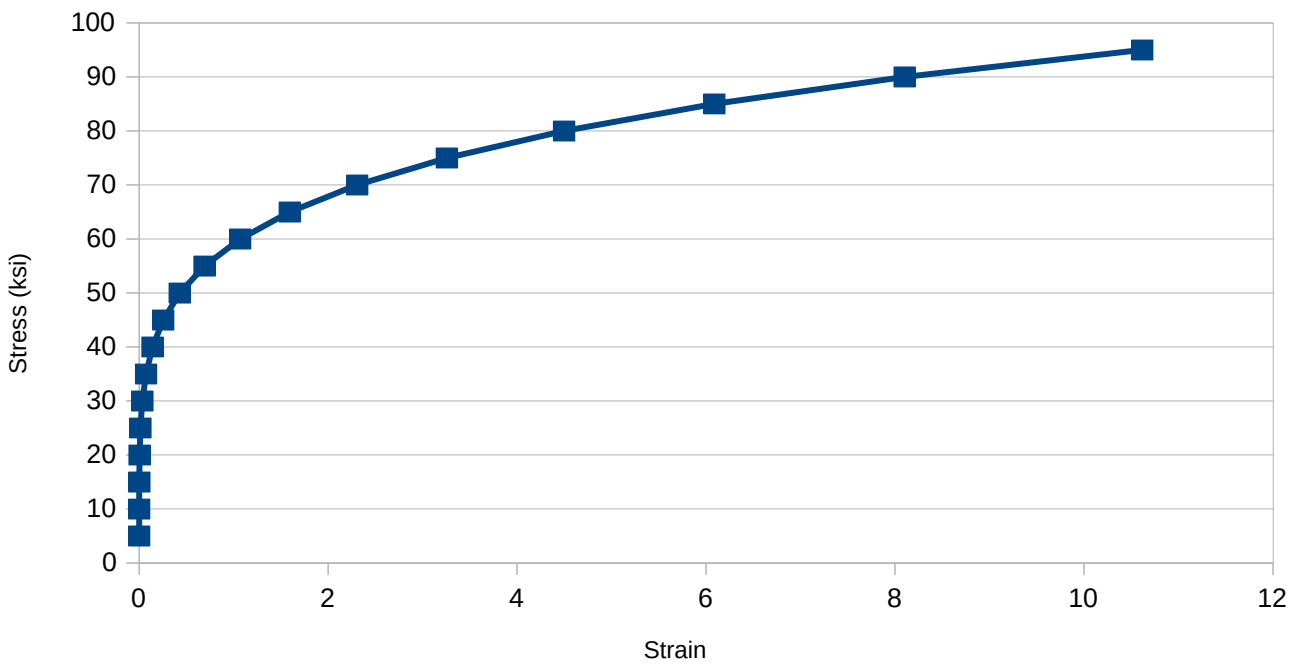
(a) $E = \frac{d\varepsilon}{d\sigma} = \left(\frac{\sigma}{30000} + \frac{\sigma}{70000}\left(\frac{\sigma}{50}\right)^5\right)' = \left[\frac{1}{30000} + \frac{1}{70000}\left(\frac{\sigma}{50}\right)^5 + \frac{\sigma}{70000}\cdot 5\left(\frac{\sigma}{50}\right)^4\right]_{\sigma=0}$

$\frac{d\varepsilon}{d\sigma} = \frac{1}{30000}$

$E = \frac{d\sigma}{d\varepsilon} = 30000ksi$

$\frac{\delta l}{.015} = 5400kss$

(b) On spreadsheet

$0.015 = \frac{\sigma}{30000}\left[1+\frac{3}{7}\left(\frac{\sigma}{50}\right)^5\right]$

5	0.0002
10	0.0005
15	0.0015
20	0.0051
25	0.0142
30	0.0343
35	0.0732
40	0.1418
45	0.2546
50	0.4302
55	0.6921
60	1.0684
65	1.5934
70	2.3073
75	3.2570
80	4.4966
85	6.0879
90	8.1011
95	10.6150